

Evaluating surface functionalization strategies in carbon nanotube-based drug delivery for cancer therapy

A review

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Abstract

Cancer is a large contributor to global mortality rates. Currently, standard chemotherapy is systemic in nature and limited by its non-specific biodistribution, causing systemic toxicity, therefore driving the urgent need for precision nanomedicine. Among emerging nanoscale delivery platforms, carbon nanotubes (CNTs) offer unique advantages due to their exceptional surface-area-to-volume ratio and anisotropic geometry, which allows them to translocate across cellular membranes. However, the inherent hydrophobicity and potential mechanical cytotoxicity of pristine CNTs present significant clinical hurdles. This review outlines the use of varied advanced surface functionalization strategies used to overcome biophysical challenges toward targeted biomedicine. Furthermore, this review postulates the therapeutic potential and remaining translational bottlenecks of surface-functionalised carbon nanotube systems in biomedicine. Finally, we briefly evaluate the potential prospects and limitations of such materials towards widespread clinical translation in the treatment of cancers.

Keywords: Carbon nanotubes, Cancer therapy, Drug Delivery, Chemotherapeutics. Nanoparticle Surface Modification

1. Introduction: Current Challenges in Oncology

Cancer is a disease that remains a large contributor to global mortality rates. Most cancer treatments today still depend on systemic chemotherapy, which spreads throughout the entire body.¹ Because this chemotherapy cannot distinguish between healthy and cancerous tissues, it causes severe side effects. This often forces doctors to limit dosages, thereby limiting complete targeting of cancer cells. Thus, a large challenge facing oncology involves finding drug delivery mechanisms that are selective toward cancerous cells to ensure limited effect toward healthy tissues.² To overcome these systemic limitations, nanoscale delivery platforms are essential. Among various nanoparticles, engineered carbon nanotubes (CNTs) are very promising due to their remarkable architectural advantages.^{4,5} Their anisotropic, high-aspect-

ratio structure may enable membrane penetration through mechanisms including endocytosis and possible nanoneedle-like insertion. As illustrated in Figure 1, this needle-like morphology offers distinct kinetic advantages over traditional spherical nanocarriers, such as conventional liposomes or polymeric nanospheres.

Specifically, the cellular internalization mechanisms of functionalized CNTs differ distinctly from those of conventional spherical nanoparticles. On the left side of Figure 1, the direct translocation pathway, often referred to as the nanoneedle mechanism, is depicted. Some studies have proposed a nanoneedle-like translocation mechanism in which CNTs may penetrate the lipid bilayer; however, endocytosis is still considered the dominant uptake pathway in many biological systems. Consequently, the attached therapeutic payload is transported directly into the cytosol, effectively avoiding the vesicular entrapment that often degrades delicate cargo. Conversely, the right side of Figure 1 illustrates the behavior of traditional isotropic, spherical nanocarriers. Due to their geometric symmetry, these spherical structures are generally unable to pierce the lipid membrane. Instead, they must rely on energy-dependent endocytic pathways, which engulf the particles into intracellular vesicles (endosomes).

This vesicular entrapment poses a major challenge, as the delivered drugs can become trapped within acidic compartments, leading to premature enzymatic degradation before reaching their intended targets. As highlighted in research reviewed by Zare et al.,⁶ this structural comparison explains why the unique high aspect ratio of cylindrical nanotubes facilitates potentially advantageous intracellular delivery kinetics. This potential "nanoneedle" insertion mechanism is hypothesized to facilitate the direct delivery of therapeutics into the cytosol.⁷ However, the precise intracellular uptake mechanism of CNTs remains under debate. While several studies propose partial membrane penetration through nanoneedle-like insertion, many experimental models suggest that endocytosis remains the dominant cellular internalization pathway under physiological conditions. Additionally, their exceptional surface-area-to-volume ratio allows for the conjugation of numerous therapeutic molecules onto a single carrier,³ enabling high localized intracellular accumulation while minimizing the required overall injected drug quantity.

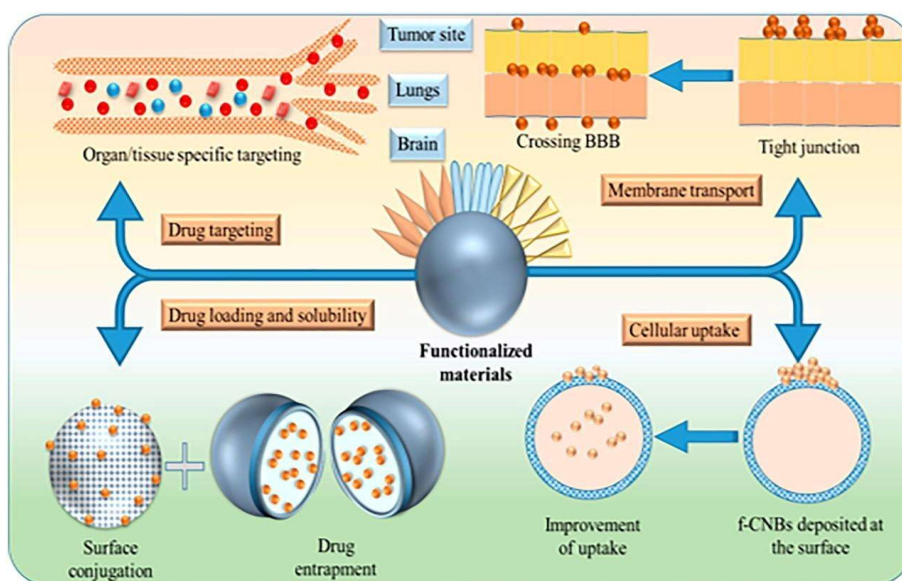


Figure 1 | Comparison of cellular internalization pathways between anisotropic carbon nanotubes and spherical nanoparticles. The high-aspect-ratio, needle-like geometry of CNTs facilitates direct perpendicular translocation across the lipid bilayer into the cytosol, bypassing the receptor-mediated restrictions faced by isotropic, spherical vehicles. Adapted from Debnath et al. ³

Despite these mechanical advantages, utilizing raw CNTs requires overcoming severe biological incompatibility.⁸ In their pristine state, unfunctionalized CNTs are hydrophobic; upon contact with the physiological environment, they aggregate rapidly and induce severe toxicity.⁹ Therefore, chemical surface functionalization is widely considered an essential prerequisite for any in vivo application.¹⁰ Nevertheless, despite promising preclinical findings, CNT-based nanomedicine remains largely experimental due to unresolved concerns regarding long-term toxicity, biodegradation, pharmacokinetic variability, and large-scale manufacturing reproducibility. By camouflaging the hydrophobic carbon surface, chemical modifications render the CNTs biocompatible and stable, ensuring homogeneous dispersion and providing the necessary anchor points for targeting ligands. To demonstrate this, pioneer experiments by Heister et al.⁷ compared unmodified carbon nanotubes directly with chemically modified variants; they observed that while raw CNTs formed large, toxic agglomerates that induced rapid cytotoxicity and reduced cell viability, nanotubes wrapped with biocompatible polymers maintained a uniform dispersion and showed negligible cytotoxicity. Similarly, Hwang et al.¹¹ highlighted that functionalization with surface groups mitigates the underlying toxic carbon framework from non-specific interactions. This crucial chemical stabilization acts as the

essential link between raw carbon materials and clinical success, ensuring the nanocarriers can navigate the complex physiological environment and reach the tumor microenvironment to execute their targeted delivery.

2. Structural Foundations, Properties, and Targeting Dynamics

The physical and chemical properties of CNTs are directly determined by their initial synthesis methodologies.¹² Researchers currently rely on three fabrication techniques: arc discharge, laser ablation, and chemical vapor deposition (CVD).^{5,13}

2.1 Arc Discharge

The arc discharge technique establishes the foundation for producing crystalline carbon lattices.¹⁴ The generation of a thermal plasma between graphite electrodes produces extreme heat, which yields nanotubes with exceptional mechanical integrity and minimal sidewall defects.¹⁴ Despite this structural perfection, the process produces substantial amorphous carbon soot and residual metal impurities.^{12,15} Dineshkumar et al. emphasize the aggressive acid washing required to purge these byproducts, which often inadvertently introduces functional defects into the pristine carbon framework.¹⁵ Consequently, this reliance on harsh post-synthesis purification severely limits the direct application of arc discharge in manufacturing ultra-pure pharmaceutical nanocarriers.

2.2 Laser Ablation

Conversely, laser ablation provides dimensional precision over the synthesized nanotubes. By utilizing a continuous laser beam to vaporize graphite targets, researchers can strictly regulate the structural geometry. Tekade et al. highlight that this precise control yields carbon tubes with highly uniform diameters, which is a critical prerequisite for achieving predictable and consistent drug loading profiles.¹⁶ Nevertheless, this precision comes at the cost of immense operational energy demands and necessitates similar complex post-synthesis purification. As a result, laser ablation remains confined to small-scale laboratory optimization rather than serving as a viable pathway for large-scale industrial manufacturing.¹⁶

2.3 Chemical Vapor Deposition (CVD)

To circumvent the limitations of thermal vaporization, chemical vapor deposition (CVD) has emerged as the standard for biomedical CNT fabrication. By depositing a transition metal

catalyst onto a substrate and introducing a carbon-rich gas within a heated chamber, CVD effectively bypasses destructive plasma environments. Hwang et al. observe that this methodology empowers researchers to customize tube length, diameter, and spatial alignment.¹¹ Extending this work, Rahamathulla et al. confirm that CVD facilitates reproducible, large-scale production batches that satisfy stringent health regulatory standards.¹⁷ Due to its scalability and controllable synthesis parameters, CVD is widely considered one of the more practical fabrication methods for biomedical CNT production.

The chosen synthesis technique directly determines the structural classification of the nanotubes, single-walled (SWCNTs) or multi-walled carbon nanotubes (MWCNTs)³, which in turn dictates their maximum payload capacity and mechanical rigidity. Table 1 delineates how this structural divergence influences the specific class of therapeutic agents each variant can optimally transport. Crucially, while layer count is a defining metric, actual *in vivo* toxicity and clinical applicability depend on a multifactorial interplay of tube length, purity, and surface chemistry.¹¹ For instance, the inherent pliability of SWCNTs renders them ideal for the electrostatic condensation of fragile genetic materials like siRNA, whereas the multilayered architecture of MWCNTs provides the mechanical stability required to transport dense chemotherapeutic loads without succumbing to hemodynamic shear stress.¹¹ Thus, matching the nanotube's physical architecture to the designated payload is essential for maximizing therapeutic efficacy.

Table 1 | Structural and Operational Parameters of SWCNTs and MWCNTs

Biophysical Parameter		Single-Walled Carbon Nanotubes (SWCNTs)	Multi-Walled Carbon Nanotubes (MWCNTs)
Fundamental Structure		A single seamless cylindrical graphene lattice	Multiple concentric nested graphene cylinders
Outer Diameter Range		0.4 – 2.0 nm ¹⁰	2 – 100 nm ¹⁰
Synthesis Catalyst		Requires transition metal catalysts	Synthesized using transition metal catalysts (though catalyst-free methods have been explored)

Initial Output Purity	Generally lower, contains residual metallic impurities	High initial purity, fewer structural sidewall defects
Mechanical Flexibility	Pliable, susceptible to bending or twisting	Rigid architecture, resistance to deformation
Optimal Payload Profile	Optimal for small molecules and genetic materials (siRNA)	Optimal for heavy metal complexes and chemotherapy loads.
Primary Toxicity Mechanisms	Higher surface-area-induced cytotoxicity; pronounced generation of reactive oxygen species (ROS). ¹⁸	Toxicity is driven by high rigidity, a prominent cause of "frustrated phagocytosis" and granulomas. ¹⁹

The chosen targeting mechanism defines the nanocarrier's journey through the body. Passive targeting strategies rely upon the Enhanced Permeability and Retention (EPR) effect.¹ To help visualize these different setups, Figure 2 visualizes the operational setups for these distinct methods

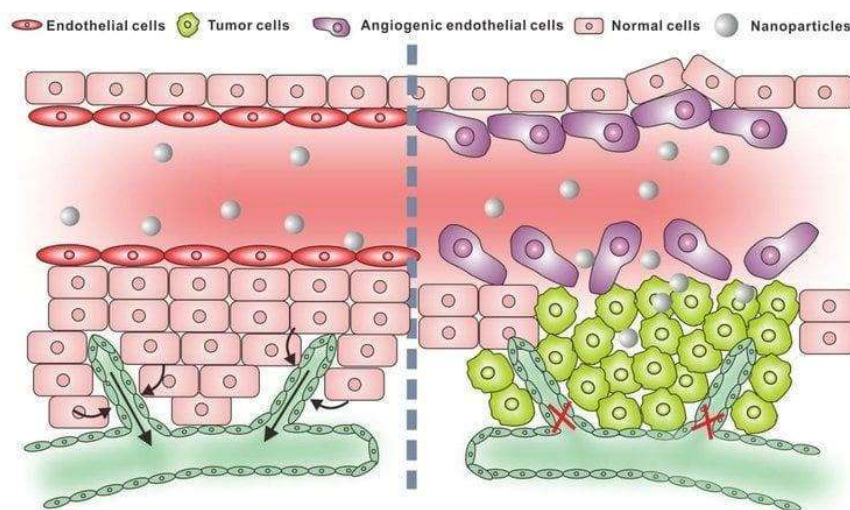


Figure 2 | Schematic representation of the Enhanced Permeability and Retention (EPR) effect. In normal tissues (left), the tight endothelial junctions prevent nanoparticle extravasation. In tumors (right), the leaky vasculature characterized by large intercellular fenestrations allows CNTs to infiltrate the interstitium. The lack of functional lymphatic drainage leads to the progressive accumulation of nanocarriers within the tumor microenvironment. Adapted from Xie et al.²⁰

Although the EPR effect has been demonstrated in murine tumor models, its translational effectiveness in human tumors remains inconsistent due to substantial interpatient variability in vascular permeability, stromal density, and lymphatic drainage efficiency. To sustain rapid proliferation, solid tumors secrete angiogenic factors that induce disorganized neovascularization. These neovessels feature compromised endothelial walls with large intercellular fenestrations.¹ As systematically illustrated in Figure 2, CNTs of appropriate dimensions extravasate through these gaps into the tumor interstitium, where an impaired lymphatic drainage system leads to their progressive entrapment and accumulation.² However, passive targeting through the EPR effect faces substantial clinical limitations. Pathologically elevated interstitial fluid pressure within solid tumors generates an outward hydrostatic gradient. This elevated IFP establishes a physical barrier that restricts deep parenchymal penetration, often leaving the hypoxic tumor core untreated.²⁰ To bypass these barriers, research has shifted toward active targeting strategies. Active targeting involves the chemical conjugation of specific navigation molecules to the external CNT surface. Aptamers represent an efficient class of targeting ligands.²⁰ Aptamers are short, artificially synthesized single-stranded oligonucleotide sequences. Through the SELEX process, they fold into complex three-dimensional spatial conformations. This geometric conformation allows aptamers to bind with high affinity to biomarker receptors overexpressed on the cancer cell membrane.²⁰

Beyond surface attachment, advanced therapeutic designs target specific internal organelles such as the mitochondria and the nucleus, which possess fortified membrane barriers.²¹ Functionalizing the CNT surface with organelle-localization signal peptides guides the therapeutic payload into these deep subcellular compartments. Releasing cytotoxic agents directly inside the mitochondria triggers immediate apoptosis.²¹ Demonstrating the efficacy of active targeting, recent studies have functionalized multi-walled carbon nanotubes (MWCNTs) with bromocriptine to specifically target dopamine receptors, which are overexpressed on certain lung cancer cell lines. This precise receptor-mediated navigation successfully triggered localized apoptosis in A549 and QU-DB lung cancer cells without inducing significant cytotoxicity in healthy MRC5 fibroblasts.²² Figure 3 illustrates this multi-stage approach, showing the transition from vascular extravasation to organelle penetration as a strategy to increase therapeutic efficacy.

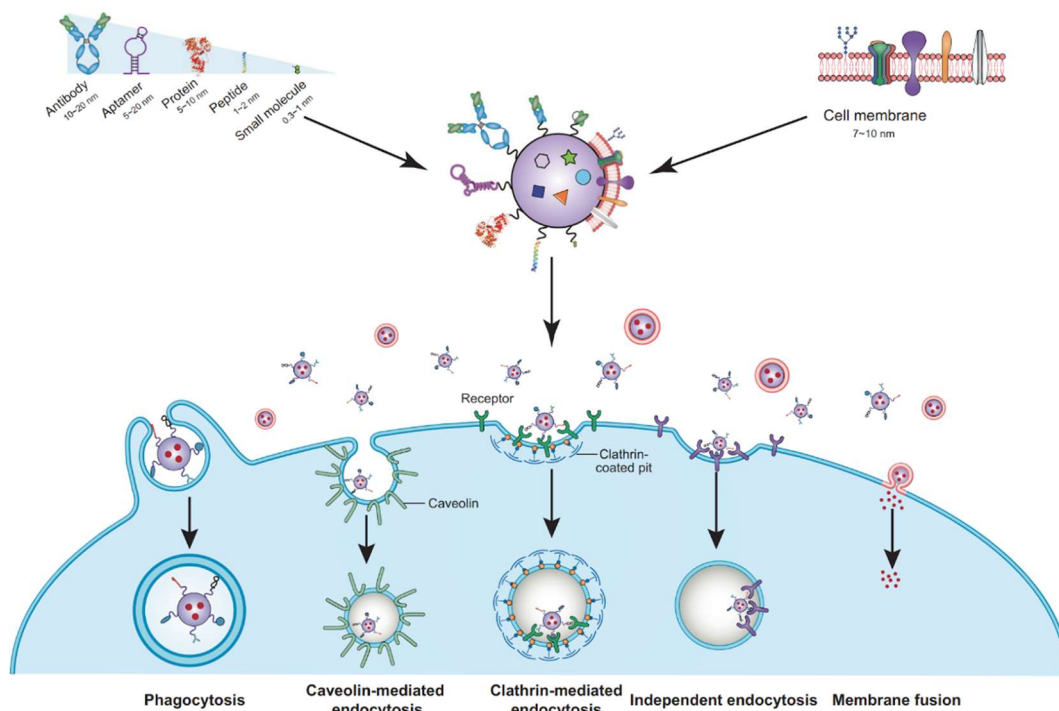


Figure 3 | Synergistic targeting dynamics integrating passive extravasation and ligand-mediated cellular internalization. The schematic demonstrates how nanocarriers, which are guided by specific surface functionalization, first exploit the EPR effect for tumor accumulation, followed by active receptor binding for endocytosis, and finally utilize specific peptides to reach deep organelles, thereby maximizing targeted cytotoxicity while sparing healthy tissues. Adapted from Fan et al.²

3. In-depth Evaluation of Surface Functionalization Strategies

3.1 Covalent Functionalization

Surface functionalization is essential to neutralize the innate toxicity of raw carbon nanomaterials and prevent in vivo agglomeration caused by van der Waals interactions. Covalent functionalization offers the most chemical methodology by establishing stable bonds directly on the carbon crystal lattice. This process follows two pathways: defect-site derivatization (Fig. 4a) and direct sidewall functionalization (Fig. 4b). Defect-site derivatization involves strong acid oxidation, using concentrated sulfuric and nitric acids, to generate structural defects and open the terminal caps.¹⁰ At these localized defect sites, reactive oxygenated functional groups, such as carboxyl (-COOH), are synthesized to act as permanent anchors.¹⁰ In comparison, direct sidewall functionalization (Fig. 4b) utilizes high-energy

chemical reactions, such as 1,3-dipolar cycloaddition, to attach functional moieties to the pristine sp^2 carbon framework without prior oxidative damage. Both pathways facilitate the subsequent conjugation of biocompatible polymers and ensure that the hydrophilic coating remains intact under hemodynamic shear stress.

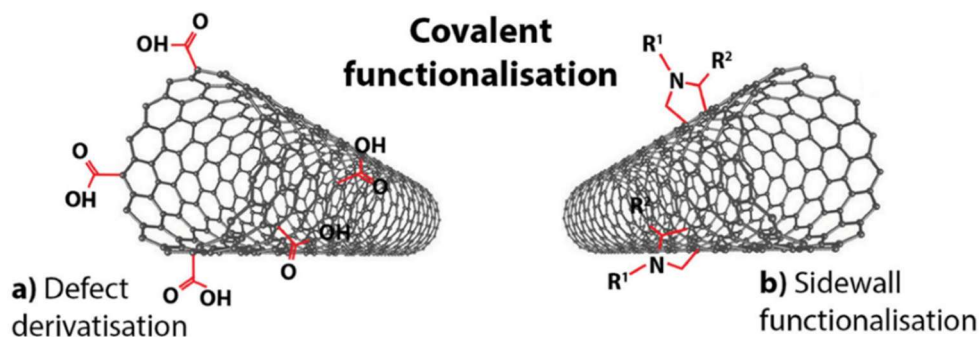


Figure 4 | Covalent modification pathways for establishing stable chemical anchors on the carbon lattice. (a) Defect-site functionalization exploits open ends or oxygenated sites, while (b) direct sidewall functionalization attaches moieties to the sp^2 carbon framework. Both pathways enhance the stability of the hydrophilic polymer shield under physiological hemodynamic shear stress. Adapted from Oprych et al.²³

Surface functionalization strategies have largely been used to introduce hydrophilicity for the biocompatibility of nanomedicines. The gold standard polymer for this purpose is polyethylene glycol (PEG).²⁸ Polymer chains are grafted onto the acid-oxidized defect sites via esterification, forming a dense, hydrophilic steric barrier surrounding the carbon tube.²² This protective shield is vital to reduce opsonization, where circulating serum proteins bind to foreign objects, marking them for clearance by the mononuclear phagocyte system.²⁴ Several studies have reported significantly reduced macrophage uptake following dense PEGylation, contributing to prolonged circulation times and reduced immune clearance.^{13,18} Saliev demonstrated that optimizing the structural length of the PEG chains effectively masks the nanotubes from opsonin proteins.¹³ Supporting this, empirical evidence from Mohanta et al. revealed that densely functionalized single-walled carbon nanotubes (SWCNTs) significantly lowered the generation of reactive oxygen species (ROS) at biointerfaces when compared to their pristine counterparts.^{13,18} By repelling these opsonins, PEGylation prolongs the in vivo circulation time, reducing elimination. Optimizing the PEGylation process requires rigorous statistical control.

Researchers often employ Response Surface Methodology (RSM) to determine the optimal reaction parameters.²⁵ Variables such as initial PEG concentration, reaction incubation time, and solvent pH directly influence the final grafting density. Statistical analyses confirm that the initial PEG concentration exerts the strongest thermodynamic impact on the substitution of free amino groups.²⁶ Yet, despite offering excellent serum stability, covalent functionalization forces a notable material trade-off. Aggressive acid oxidation and direct covalent bonding disrupt the continuous sp^2 hybridized carbon network.¹⁰ This disruption diminishes the mechanical tensile strength and natural electrical conductivity of the tubes. This loss limits their application in advanced electrical biosensing, though they remain effective as robust, long-circulating chemotherapeutic delivery vehicles. To demonstrate this *in vivo* efficacy, Liu et al. showed that PEGylated SWCNTs loaded with paclitaxel (PTX) achieved higher tumor suppression in murine breast cancer models compared to clinical Taxol, validating the importance of the polymer shield.²⁷

3.2 Non-Covalent Functionalization

Unlike oxidative methods, non-covalent functionalization preserves the pristine carbon crystal lattice by utilizing physical interactions rather than breaking chemical bonds. These supramolecular strategies, including polymer wrapping, endohedral encapsulation, and π - π stacking, maintain the sp^2 hybridized network and high electrical conductivity. Polymer wrapping provides a prevalent non-covalent functionalization approach, as highlighted in Figure 4. Amphiphilic block copolymers coil around the circumference of the nanotubes. Phospholipid-polyethylene glycol (PL-PEG) serves as a regarded amphiphilic wrapping agent in preclinical trials.²⁸ The hydrophobic lipid tail of the PL-PEG molecule interacts with the non-polar carbon surface via hydrophobic forces, while the hydrophilic PEG head points outward into the surrounding aqueous solution.²⁹ This polymeric shield maintains the internal electronic structural integrity of the tube while ensuring high colloidal water solubility on the exterior. By maintaining this colloidal stability in physiological fluids, the non-covalent polymer shield prolongs circulation time by delaying immune clearance, ensuring they circulate long enough to deeply penetrate the tumor microenvironment. In addition, the polymer wrapping method prevents cellular aggregation without introducing defects into the electronic band structure. The preservation of these conductive properties makes non-covalently wrapped

CNTs a promising platform for multimodal photothermal combination therapies, allowing for precise, light-triggered cancer cell eradication.

Another key mechanism is π - π stacking, where molecules with conjugated aromatic rings interact directly with the CNT surface.³¹ A classic application is the loading of Doxorubicin (DOX), which adsorbs onto the hydrophobic graphene surface via its flat, tetracyclic aromatic structure. As modeled in Figure 5, this orientation enables high-density drug loading without chemical cross-linkers.³¹ To optimize this physical interaction force, computational chemistry simulations are deployed. Density Functional Theory (DFT) calculations compute the precise solvation free energies and binding affinities.³² These sophisticated simulations demonstrate that bioactive dipeptides, such as carnosine, bind to the sidewalls of CNTs in physiological aqueous environments without detaching prematurely.³² When evaluating these two paradigms for immediate clinical translation, covalent functionalization presents a more predictable and proven pathway for human application. While non-covalent methods offer a way to preserve the intrinsic physical properties of the nanotube, they lack the long-term structural "ruggedness" required to survive the complex, high-shear environment of the human circulatory system. Because regulatory bodies prioritize predictable pharmacokinetic stability, as evidenced by the clinical success of covalently PEGylated lipid nanoparticles, the irreversible nature of covalent bonds provides the most reliable path to ensuring the nanocarriers reach the tumor site intact.

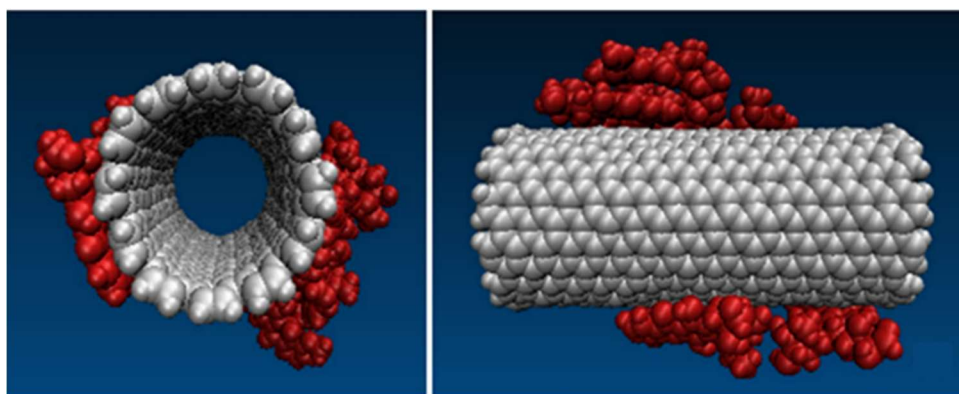


Figure 5 | Molecular orientation of doxorubicin facilitating high-density non-covalent payload loading. The model illustrates how doxorubicin (DOX, shown in red) naturally adsorbs onto the carbon nanotube surface through strong π - π stacking interactions, allowing for efficient drug encapsulation without the need for complex chemical cross-linkers. Adapted from Arabian et al.³⁰

Overall, covalent and non-covalent functionalization strategies present distinct advantages and limitations. Covalent approaches generally provide greater structural stability and stronger ligand retention under physiological conditions, though they may disrupt the intrinsic electronic and mechanical properties of CNTs through modification of the sp^2 carbon lattice. In contrast, non-covalent strategies better preserve the native structural and conductive properties of CNTs however, the weaker intermolecular interactions may reduce long-term coating stability during systemic circulation. Consequently, selecting an optimal functionalization strategy requires balancing structural stability, biocompatibility, and preservation of nanotube physicochemical properties according to the intended therapeutic application.

4. Advanced Therapeutic Payload Delivery Systems

4.1 Small-Molecule Chemotherapeutics and Heavy Metals

Functionalized CNTs operate as secure molecular transport vehicles for a diverse array of therapeutic payloads. In their unencapsulated free form, small-molecule chemotherapeutics face rapid renal clearance. They distribute non-specifically, causing severe toxicity in healthy non-target organs.¹ CNTs encapsulate or adsorb these cytotoxic molecules to alter their systemic pharmacokinetics. The targeted delivery of Ixazomib exemplifies this nanomedical application. Ixazomib operates as a potent, reversible proteasome inhibitor utilized in multiple myeloma management.²⁴ Administering free Ixazomib causes diffuse systemic toxicity. To mitigate this, researchers covalently encapsulated Ixazomib molecules inside the internal cavity of PEG-wrapped MWCNTs.²⁴

This engineered MWCNTs-PEG-Ixazomib complex targets RPMI 8226 myeloma cells with high precision. The carbon transport system leverages receptor-mediated endocytosis to penetrate the plasma membrane of the malignant plasma cell.²⁴ In vitro cytotoxicity assays demonstrate that this system maintains high apoptotic efficacy in destroying abnormal myeloma cells. At the same time, the stable encapsulation reduces cytotoxicity to the surrounding healthy bone marrow stromal cells.²⁴

Providing a preclinical model, a seminal in vivo study by Liu et al. demonstrated that loading Doxorubicin (DOX) onto PEGylated SWCNTs prolonged the drug's blood circulation time.²⁷ In murine xenograft models, this functionalized system achieved a higher tumor growth inhibition

rate while effectively mitigating the severe cardiotoxicity and mortality associated with free DOX administration.

Despite these promising outcomes, the majority of CNT-based chemotherapeutic systems remain confined to preclinical animal studies. Their long-term biodistribution, biodegradation kinetics, and potential chronic toxicity in humans remain insufficiently characterized.

Materials scientists also integrate heavy transition metal complexes onto carbon platforms. For example, functionalized MWCNTs frequently serve as foundational substrates to deliver heavy transition metal complexes, such as platinum-based agents. These carbon-based nanocomposites report up to a 3-fold higher tumor uptake compared to free cisplatin, significantly enhancing targeted accumulation while avoiding severe systemic toxicity.

These examples demonstrate that functionalized carbon platforms improve the pharmacokinetic profiles of both small-molecule and heavy-metal chemotherapeutics. Beyond conventional synthetic drugs, functionalized CNTs are emerging as potent vectors for natural anti-cancer agents. Modifying CNTs with carboxylic (-COOH) or amine (-NH₂) groups, combined with a chitosan and stearic acid coating, has been shown to significantly enhance the cellular uptake of natural compounds like Ferulic acid and Diosgenin in breast (MCF-7) and liver (HepG2) cancer cells.³⁵ Encapsulating standard chemotherapeutics like Oxaliplatin within MWCNT-PEG systems via nano-extraction effectively prolongs drug release. This specific formulation dramatically increases selective cytotoxicity against HT29 colorectal cancer cells through the enhanced activation of Pt-DNA adducts and γ -H2AX phosphorylation.³⁶

4.2 Systemic Delivery of Genetic Material (siRNA)

Beyond small-molecule chemotherapeutics, CNTs can also encapsulate and protect fragile genetic materials. Small interfering RNA (siRNA) represents a promising biological tool capable of silencing specific oncogenes. However, naked siRNA degrades very rapidly in the systemic bloodstream. Circulating endogenous RNase enzymes cleave the phosphodiester backbone of RNA within minutes of intravenous injection.²⁹ Delivering siRNA safely to the tumor requires a physical protective vehicle. SWCNTs non-covalently functionalized with PL-PEG perform this protective task well.²⁸

Naked siRNA carries a very strong polyanionic negative charge due to its phosphate groups. Researchers modified the terminal ends of the PL-PEG layer with primary amine groups. This modification provides a positively charged, polycationic surface.²⁸ The positively charged SWCNTs attract, condense, and bind the negatively charged siRNA via strong electrostatic interactions.²⁸ This resulting SWNT-siRNA nanoplex penetrates cancer cells primarily through energy-dependent endocytosis pathways.²⁸

Once engulfed inside the endosome, the carrier must execute an escape maneuver to avoid lysosomal acid degradation. The endosomal escape is frequently attributed to the 'proton sponge' hypothesis, though recent literature also highlights direct polymer-lipid membrane destabilization as a contributing mechanism.³⁷ As the vesicle's pumps attempt to acidify the internal environment, the amine groups on the CNT surface act as a weak base buffer, absorbing a massive influx of protons. To maintain electrical neutrality, chloride ions follow the protons inside.³⁷ This rapid accumulation of ions increases the internal osmotic pressure. Consequently, water influx causes endosomal swelling, causing it to swell and rupture, thereby releasing the intact siRNA directly into the cellular cytoplasm.^{28,29} Within the cytosol, the siRNA sequence integrates into a multiprotein complex known as the RNA-induced silencing complex (RISC).²⁹ The Argonaute protein within this RISC complex uses the siRNA as a guide to scan the cell. By catalytically degrading specific complementary mutant mRNAs, this targeted cleavage process silences oncogenes, thereby reducing the expression of proteins associated with tumor survival and proliferation.

Demonstrating this in vivo, Bartholomeusz et al. successfully delivered siRNA targeting the HIF-1 α oncogene via CNTs, achieving specific gene silencing and arresting tumor growth.³⁸ Complementing this, Liu et al. utilized CNTs to deliver siRNA against the CXCR4 receptor, reducing breast cancer metastasis to the lungs.³⁹ Collectively, these empirical findings demonstrate that functionalized CNTs provide a nuclease-resistant platform capable of translating fragile genetic concepts into highly promising systemic therapeutic candidates.

5. Stimuli-Responsive "Smart" Release Mechanisms

5.1 Endogenous Triggers: Exploiting pH and Enzymes

Precise spatiotemporal control of drug release is crucial for optimizing overall clinical safety. Stimuli-responsive mechanisms allow functionalized CNTs to be programmed. They release

their cytotoxic drugs at the required time and spatial location.⁴⁰ Endogenous triggers exploit the unique biochemical abnormalities intrinsic to the tumor microenvironment (TME). Solid tumors exhibit high rates of aerobic glycolysis. This metabolic shift is known as the Warburg effect.⁴¹ This inefficient metabolic process continuously produces and exports large quantities of lactic acid. Thus, the extracellular pH of the tumor interstitium drops to an acidic range of around 5.5–6.8.⁴⁰

pH-sensitive CNTs utilize specific acid-labile chemical bonds to tether the drug to the carbon surface. Hydrazone and ester linkers are frequently employed for this purpose.⁴⁰ The carbon carrier stably retains the payload while circulating in healthy, neutral-pH blood. Once the nanocarrier extravasates into the acidic tumor microenvironment, these hydrazone bonds undergo rapid hydrolysis. A functionalized CNT-based complex demonstrates this targeted release mechanism.^{33,42} The cumulative release rate of DOX from this specific platform spikes under acidic in vitro conditions. This chemical response ensures the largest possible drug volume is released within the tumor microenvironment.³³

Enzyme-triggered systems provide another sophisticated endogenous release mechanism. To degrade the extracellular matrix and invade surrounding healthy tissues, metastatic tumors frequently overexpress specific proteolytic enzymes. Matrix metalloproteinases (MMPs) and cathepsins are the most prominent examples of these overexpressed proteases.⁴⁰ To exploit this biological vulnerability, researchers synthesize specially designed short peptide sequences that serve as sacrificial substrates. For instance, the specific hexapeptide sequence PLGLAG (Proline-Leucine-Glycine-Leucine-Alanine-Glycine) is engineered as a structural bridge linking the cytotoxic drug to the CNT surface.⁴⁰ This particular sequence is sensitive to MMP-2, an enzyme upregulated in aggressive carcinomas. While circulating through the healthy bloodstream where MMP-2 expression is negligible, the specific peptide bonds remain intact, keeping the cytotoxic payload safely sequestered. Upon reaching the tumor site, concentrated MMPs recognize and sever the PLGLAG linker (e.g., between the glycine and leucine residues), ensuring a highly localized burst release. This specific biochemical cleavage concentrates the maximum lethal dosage at the lesion site while minimizing systemic exposure to off-target tissues. It concentrates the maximum lethal dosage at the lesion site while minimizing systemic exposure to off-target tissues.

Empirical data supports these mechanisms: Gu et al. observed that hydrazone-linked DOX released over 75% of its payload within 48 hours at pH 5.5, compared to less than 15% at pH 7.4.⁴³ Yao et al. demonstrated that PLGLAG-cleavable CNTs released their payload in MMP-rich tumor homogenates, leaving normal tissue cultures unharmed. These controlled release profiles confirm that endogenous triggers successfully restrict drug cytotoxicity to the tumor microenvironment. Advancing beyond standard nanotubes, emerging platforms utilizing mesoporous carbon nanomaterials offer exceptionally large surface areas engineered for dual-triggered smart release. These intelligent systems are designed to remain highly stable in blood circulation but rapidly disintegrate and release their payloads upon encountering both the acidic pH of the tumor lysosome and localized redox state variations, thereby maximizing tumor-specific cytotoxicity.^{44,45}

5.2 Exogenous Triggers: Thermal and Near-Infrared (NIR) Responses

Exogenous triggers leverage targeted physical stimuli applied intentionally from outside the patient's body. Pristine CNTs possess powerful optical absorption properties based on their unique electronic band structures. They absorb near-infrared (NIR) light in the biological windows (700-1100 nm).⁴⁶ Unlike visible or ultraviolet light, NIR light safely penetrates deep into human tissue without causing surface thermal burns or significant photon scattering.⁴⁶

As illustrated in Figure 6, when illuminated by an external, focused NIR laser beam, CNTs accumulated within the tumor absorb this specific optical energy. Through non-radiative relaxation pathways, they convert the absorbed photons into significant localized heat.⁴⁷ This thermogenic reaction serves an effective dual therapeutic purpose. First, the sudden intense heat melts the surrounding thermo-sensitive polymer coatings. This phase change triggers a rapid burst release of the encapsulated drug.^{40,48} Second, the induced hyperthermia (reaching temperatures above 42°C) directly denatures cellular proteins. It induces localized thermal ablation, leading to rapid coagulation necrosis.

Nevertheless, several technical limitations remain associated with photothermal therapy. Uneven heat distribution, limited penetration depth in large tumor masses, and the risk of off-target thermal injury may compromise treatment uniformity and safety in deep tissues.

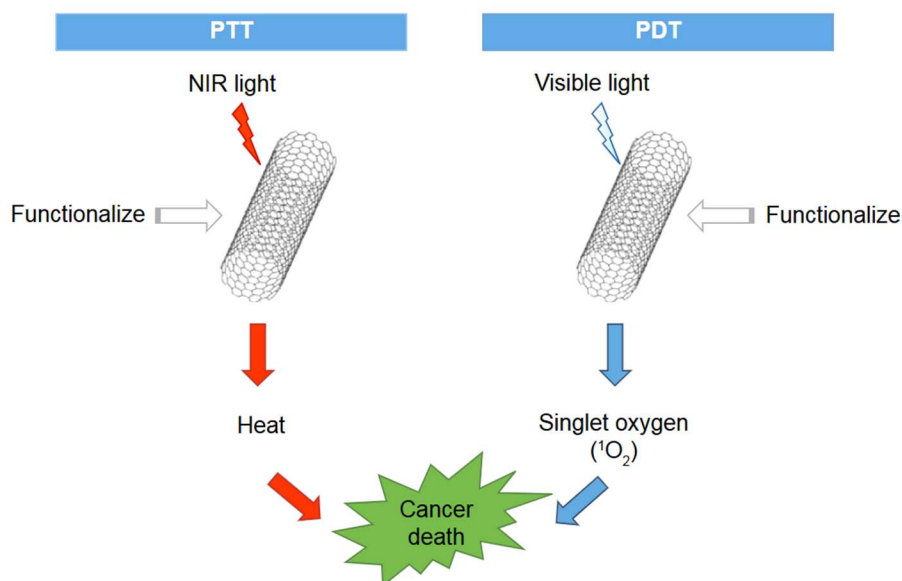


Figure 6 | Light-triggered exogenous pathways for synergistic thermal and oxidative cancer cell eradication. Functionalized CNTs utilize near-infrared (NIR) light to generate localized heat for Photothermal Therapy (PTT) or visible light to produce cytotoxic singlet oxygen for Photodynamic Therapy (PDT), offering clinicians precise spatiotemporal control over tumor destruction. Adapted from Son et al.⁴⁹

Figure 6 demonstrates that CNTs can also facilitate Photodynamic Therapy (PDT) when functionalized and exposed to specific wavelengths of light. In this pathway, the nanocarrier aids in the generation of reactive singlet oxygen ($^1\text{O}_2$), which induces oxidative damage to malignant cells. A multifunctional SWCNT system combining chemotherapeutic drug delivery with these light-activated responses has demonstrated an accelerated rate of cancer cell destruction in murine models⁴⁷. This synergistic approach exhausts the cancer cells' heat-shock recovery capabilities while minimizing the amount of toxic drug required.

While both endogenous and exogenous triggers demonstrate remarkable efficacy in vitro, it can be argued that exogenous triggers, particularly NIR-induced photothermal responses, may offer greater external control over release kinetics and treatment timing compared to endogenous trigger systems. Endogenous systems rely heavily on the natural heterogeneity of the tumor microenvironment; fluctuations in local pH or MMP expression can lead to unpredictable drug release rates. In contrast, as shown in the simplified clinical workflow of Figure 6, exogenous NIR triggers place temporal control directly in the hands of the clinician, allowing for a precise, on-demand burst release that is significantly less dependent on the

patient's variable internal biochemistry. However, due to the inherent tissue penetration limits of NIR light, this external spatial control is currently most viable for superficial or endoscopically accessible tumors.

In clinical models, Robinson et al. utilized 808 nm NIR irradiation on PEGylated SWCNTs to raise tumor core temperatures to 50°C, achieving complete ablation in murine models.⁵⁰ This finding confirms that exogenous physical triggers provide clinicians with enhanced, on-demand spatiotemporal control over tumor destruction.

6. Overcoming Multidrug Resistance (MDR) and Modulating the Immune Microenvironment

Multidrug resistance (MDR) represents the primary physiological cause of failure in long-term clinical chemotherapy protocols. Over successive treatment cycles, cancer cells undergo genetic selection to overexpress P-glycoprotein (P-gp) efflux pumps on their outer plasma membranes.²⁸ These transmembrane ATP-binding cassette proteins act as efficient efflux mechanisms. They hydrolyze ATP to pump small-molecule chemotherapy drugs out of the cytoplasm before the drug can exert DNA damage. Free chemotherapy drugs typically enter the cell via simple passive diffusion across the lipid bilayer. This slow diffusion makes them detectable and expellable by the active P-gp pumps.

Functionalized CNTs bypass this sophisticated protein defense system by fundamentally changing the cellular entry mechanism.²⁸ CNTs enter cells through vesicular endocytosis rather than passive diffusion. They carry their therapeutic payload hidden deep inside impermeable transport vesicles. The physical presence of the P-gp pump is unable to recognize, bind, or obstruct this massive, encapsulated carbon structure.²⁸ Extensive in vitro studies show that SWNT-siRNA complexes accumulate inside drug-resistant cellular variants, regardless of how high the cell's P-gp expression levels are elevated.²⁸

Co-delivery strategies maximize the therapeutic efficacy against refractory tumors driven by genetic drug resistance. To overcome resistance caused by genomic mutations that prevent traditional drugs from binding to their targets, researchers can exploit the massive available surface area of elongated CNTs to co-deliver multiple distinct therapeutic agents. Carbon nanoparticles can also deliver a potent cytotoxic drug and a specific siRNA sequence on the

same platform.²⁹ The siRNA sequence acts as a targeted silencing agent. It silences the mRNA transcripts responsible for conferring drug resistance. For example, siRNA targeting the MDR1 gene halts the ribosomal translation of new P-gp efflux pumps²⁹. Once the targeted silencing agent downregulates the resistance mechanisms, the released chemotherapy drug easily accumulates and destroys the cancer cell. This co-delivery strategy may partially restore chemosensitivity in resistant tumor models.⁴⁰

However, multifunctional co-delivery systems substantially increase formulation complexity. Achieving reproducible loading ratios, controlled release kinetics, and scalable manufacturing remains a significant translational challenge. Despite these manufacturing challenges, co-delivery strategies offer profound clinical benefits. A prominent example is the simultaneous loading of Curcumin and Methotrexate onto an albumin-coated, functionalized MWCNT platform. This synergistic co-delivery approach amplifies active targeting capabilities and effectively bypasses traditional single-drug resistance pathways, leading to enhanced apoptosis in malignant cells while significantly mitigating off-target side effects.⁵¹

For example, Li et al. demonstrated that targeted CNTs delivered doxorubicin directly to resistant MCF-7/ADR breast cancer cells, effectively overwhelming the P-glycoprotein efflux pumps and restoring drug sensitivity.⁵²

In addition, nanomedicine currently plays a pivotal role in augmenting modern cancer immunotherapy. Solid tumors construct a dense, immunosuppressive microenvironment to actively evade destruction by cytotoxic T lymphocytes.⁵³ Nanoparticles possess the unique capability to biologically reprogram this hostile environment. Combination therapies utilize CNTs to deliver specific chemotherapy drugs capable of inducing Immunogenic Cell Death (ICD).⁵³ The ICD process ruptures the cancer cell, releasing numerous damage-associated molecular patterns (DAMPs) and hidden tumor antigens into the extracellular environment. Meanwhile, the nanocarrier releases co-loaded immune adjuvants to awaken, recruit, and stimulate antigen-presenting dendritic cells.⁵³ This engineered combination enhances the body's natural systemic anti-tumor immune response. It transforms immunologically "cold", drug-resistant tumors into highly vulnerable, inflamed targets for the patient's immune system.⁵³ As systematically summarized in Table 2, these diverse CNT-based resolution

strategies overcome the primary biological and physical barriers that cause conventional therapies to fail.

For example, Fadel et al. used CNT platforms to present antigens and deliver immune adjuvants, stimulating an expansion of tumor-specific T cells.⁵⁴ Thus, it can be claimed that CNTs serve as potent immunomodulatory vehicles capable of modulating tumor immunosuppression.

Table 2 | Mechanisms for overcoming biological tumor barriers utilizing CNT platforms.

Biological Tumor Barrier	Cellular Resistance Mechanism	CNT-Based Resolution Strategy
P-gp Efflux Pumps	Extrudes small-molecule drugs utilizing ATP hydrolysis	Bypasses the membrane pump via vesicular endocytosis
Genetic Drug Resistance	Genomic mutations prevent traditional drugs from binding	Co-delivery of targeted siRNA to silence specific mutated transcripts
Dense Tumor Stroma	High interstitial pressure restricts fluid diffusion	Surface functionalization with heavy metals to enhance therapeutic activity or imaging contrast
Immunosuppression	Deactivates cytotoxic T cells within the TME	Co-delivery of immune adjuvants to trigger Immunogenic Cell Death (ICD)

7. Toxicity, Biotransformation, Clinical Perspectives, and Regulatory Barriers

The successful translation of CNTs from the laboratory into clinical applications depends upon understanding their complex systemic toxicity profiles. In their raw, pristine state, CNTs cause severe, acute cytotoxicity at biological cellular interfaces¹⁸. The sharp physical edges of the rigid tube structure puncture fragile cell membranes. Bare carbon tubes trigger massive intracellular oxidative stress by continuously generating copious amounts of reactive oxygen species (ROS).¹⁸ These uncontrolled ROS attack lipid membranes. They cause rapid lipid peroxidation that destroys mitochondrial membranes and induces necrosis.

Reported CNT toxicity profiles vary across studies due to differences in tube length, aggregation state, surface chemistry, dosage, purity, and experimental model. Consequently, cross-study comparisons remain difficult, and establishing widely accepted safety standards for CNT-based therapeutics continues to present a major regulatory challenge.

In addition, the geometric size and shape of the carbon tube determine the innate immune system's response. Long CNTs physically mimic the dangerous pathogenic structure of asbestos fibers. Alveolar macrophages attempt to phagocytose these elongated structures but fail to internalize them.¹⁹ This process is termed “frustrated phagocytosis,” which causes the stressed macrophages to leak destructive digestive enzymes and cytokines into the surrounding tissue. This leakage triggers severe chronic inflammation, granuloma formation, and eventual tissue fibrosis.¹⁹

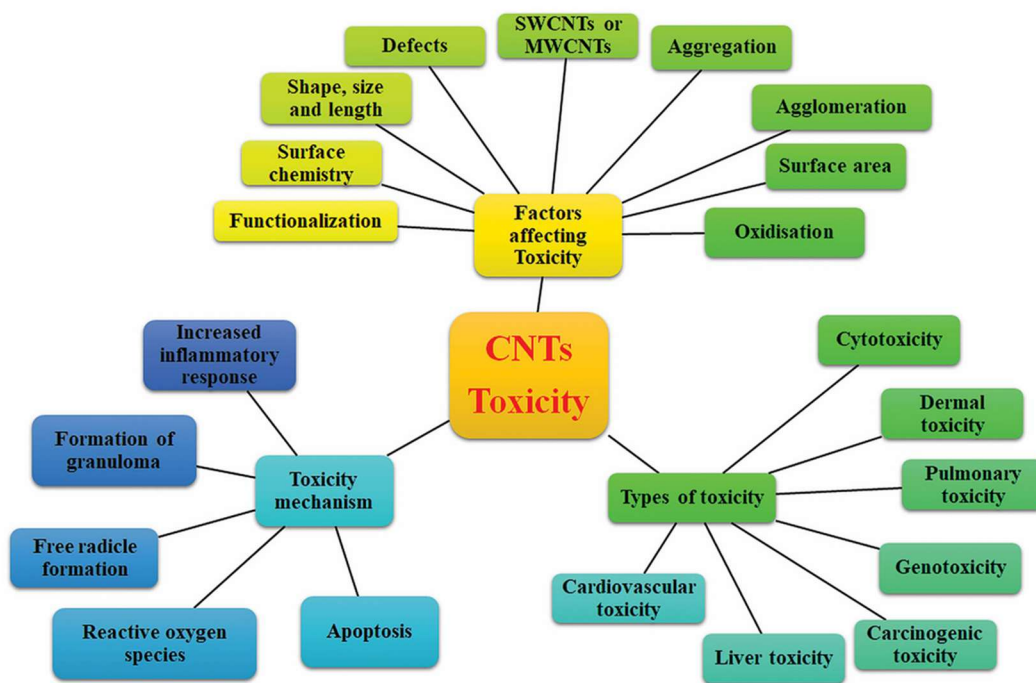


Figure 9 | Interplay between physicochemical properties and biological mechanisms in CNT-induced cytotoxicity. The schematic highlights how factors like tube length and surface chemistry drive biological damage, such as ROS generation and frustrated phagocytosis, underscoring the necessity of rigorous surface functionalization to guarantee clinical safety. Adapted from Prajapati et al.¹⁹

To mitigate these severe toxicological effects, stringent surface chemical modification is very essential. Coating CNTs with hydrophilic polymers like PEG significantly attenuates the rapid formation of the "protein corona".²⁸ The protein corona is a dense layer of circulating serum

proteins that coats all foreign objects entering the bloodstream. This hydrophilic polymer shield physically restricts specific opsonin proteins from attaching to the underlying carbon.¹⁹ As a result, it reduces recognition and clearance by the reticuloendothelial system. In addition, materials scientists now control the longitudinal tube length during the initial acid synthesis phase. Deliberately cutting the carbon tubes below the 200-nanometer threshold substantially reduces the onset of frustrated phagocytosis.¹⁸

Furthermore, significant progress has been made in addressing the potential for permanent bioaccumulation. Pioneering research demonstrates that human myeloperoxidase (hMPO) can biodegrade functionalized CNTs.^{19,55} Activated neutrophils naturally secrete hMPO during inflammatory responses. Through the generation of potent reactive radical intermediates and hypochlorous acid (HOCl), this enzyme catalyzes powerful oxidative degradation. This radical-driven mechanism steadily cleaves the sp^2 carbon network into **biocompatible**, water-soluble carbonaceous fragments, including short-chain carboxylated alkanes and alkenes.⁵⁵ In vivo toxicological assessments confirm that these biodegraded nanotubes do not elicit a pro-inflammatory pulmonary response, nor do they induce tissue granuloma formation when introduced into the lung.⁵⁵ These ultra-small degraded fragments are subsequently filtered through the renal glomerular system and excreted in urine.

Moving beyond simple targeted therapeutics, CNTs are expanding into the field of in vivo diagnostics. Carbon-based quantum dots and heavily functionalized nanotubes possess powerful, tunable photoluminescent properties.⁵⁶ They serve as ultra-sensitive, non-bleaching tracking tools inside the living body. CNTs can detect circulating tumor cells (CTCs) and specific oncogenic biomarkers even at extremely low concentrations in the blood.⁵⁷ Integrating advanced diagnostics and targeted therapeutics onto a single CNT platform enables theranostic applications.⁵⁶ Theranostic CNTs allow clinicians to track the exact spatial location and timing of drug release. Utilizing Raman spectroscopy or NIR imaging, physicians can observe tumor mass shrinkage non-invasively in real-time, allowing for immediate dosage adjustments.⁵⁸ Modern theranostic designs are increasingly integrating these carbon platforms with metallic nanoparticles, such as gold, silver, or magnetic seed structures. These multidimensional hybrid systems not only provide superior contrast enhancement for CT and MRI diagnostic imaging

but also serve as highly efficient multimodal mediators for magnetic-guided navigation and synergistic photothermal therapy.^{34,59}

Despite these promising in vitro and in vivo outcomes in preclinical laboratory settings, the commercial manufacturing challenges remain substantial. Securing approval from international health regulatory agencies requires strict, uncompromising adherence to Good Manufacturing Practice (GMP) standards.⁵⁷ Producing multifunctional, uniform industrial batches of CNTs poses significant manufacturing and scale-up challenges. A minuscule molecular variation in the average tube length or a slight deviation in PEG functional group density alters the drug's entire pharmacokinetic and biodistribution profile in human patients³.

Currently, global regulatory bodies such as the FDA and EMA, lack standardized evaluation protocols tailored specifically for non-spherical carbon nanomaterials.⁵⁷ Because of this, advancing to Phase I trials still depends on time-consuming chronic toxicity evaluations in large mammalian models. However, this regulatory gap will push the field toward a specialized, nano-specific regulatory pathway.⁵⁷ This evolution will mandate a "Characterization-by-Design" approach, requiring exhaustive data on tube length distribution and functional group density.³ In addition, given their unique physical and genetic targeting potential, CNTs offer enough clinical promise to overcome these translational bottlenecks, paving the way for more personalized and genetically synchronized nanomedicine in oncology.^{13,46,57}

Beyond toxicological concerns, the intrinsic heterogeneity of CNT synthesis remains a substantial barrier to clinical translation. Minor variations in tube diameter, chirality, surface functionalization density, and residual catalyst contamination may significantly alter biological interactions and therapeutic performance. Therefore, future progress in CNT nanomedicine will depend heavily on developing standardized synthesis and characterization protocols.

Importantly, despite decades of research, CNT-based oncology systems have not yet achieved widespread clinical approval. This gap reflects persistent concerns regarding long-term pharmacokinetics, standardized toxicological evaluation, and large-scale reproducibility.

In summary, surface functionalization strategies have substantially improved the biomedical applicability of CNTs by enhancing colloidal stability, reducing toxicity, and enabling targeted delivery. Nevertheless, significant translational barriers remain unresolved, particularly

regarding long-term biosafety, manufacturing reproducibility, and regulatory standardization. Future progress will likely depend not only on advances in nanomaterial engineering, but also on the development of rigorous characterization frameworks capable of supporting safe clinical implementation.

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